TASK-3 : Website Traffic Analysis

A. Import Libraries

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

B. Load Dataset

In [2]:
```
df = pd.read_csv("traffic.csv")
df.head()
```

Out[2]:

	event	date	country	city	artist	album	track	isrc	
0	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d8997b6-1c5fb9cc
1	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d8997b6-1c5fb9cc
2	click	2021-08-21	India	Ludhiana	Reyanna Maria	So Pretty	So Pretty	USUM72100871	23199cf5-34965c3
3	click	2021-08-21	France	Unknown	Simone & Simaria, Sebastian Yatra	No Llores Más	No Llores Más	BRUM72003904	35574e49-08a960f
4	click	2021-08-21	Maldives	Malé	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d8997b6-1c5fb9cc

In [3]:
```
df.info()
df.describe()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 226278 entries, 0 to 226277
Data columns (total 9 columns):
 #   Column   Non-Null Count   Dtype
---  ------   --------------   -----
 0   event    226278 non-null  str
 1   date     226278 non-null  str
 2   country  226267 non-null  str
 3   city     226267 non-null  str
 4   artist   226241 non-null  str
 5   album    226273 non-null  str
 6   track    226273 non-null  str
 7   isrc     219157 non-null  str
 8   linkid   226278 non-null  str
dtypes: str(9)
memory usage: 15.5 MB
```

Out[3]:

	event	date	country	city	artist	album	track	isrc
count	226278	226278	226267	226267	226241	226273	226273	219157
unique	3	7	211	11993	2419	3254	3562	709
top	pageview	2021-08-19	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741
freq	142015	35361	47334	22791	40841	40841	40841	40841

C. Data Cleaning

1. Check Missing Value

In [4]:
```
df.isnull().sum()
```

Out[4]:
```
event         0
date          0
country      11
city         11
artist       37
album         5
track         5
isrc       7121
linkid        0
dtype: int64
```

In [5]:
```
df = df.dropna()
```

2. Convert Date Columns to Datetime

In [6]:
```
df['date'] = pd.to_datetime(df['date'])
```

3. Remove Duplicate Records

In [7]:
```
df = df.drop_duplicates()
```

D. Basic Traffic Metrics

1. Total Users

In [8]:
```
users = df[['city','country']].drop_duplicates().shape[0]
print("Total Users:", users)
```

```
Total Users: 12155
```

2. Total Sessions

In [9]:
```
sessions = df.shape[0]
print("Total Sessions:", sessions)
```

```
Total Sessions: 116257
```

3. Average Session Duration

In [10]:
```
session_per_day = df.groupby('date').size()

avg_session = session_per_day.mean()

print("Average Sessions Per Day:", avg_session)
```

```
Average Sessions Per Day: 16608.14285714286
```

4. Bounce Rate

In [11]:
```
clicks_per_user = df.groupby(['city','country']).size()

bounced = (clicks_per_user == 1).sum()

bounce_rate = bounced / len(clicks_per_user) * 100

print("Bounce Rate:", bounce_rate,"%")
```

```
Bounce Rate: 27.27272727272727 %
```

E. Top Landing Pages

In [12]:
```
top_pages = df['linkid'].value_counts().head(10)

print(top_pages)
```

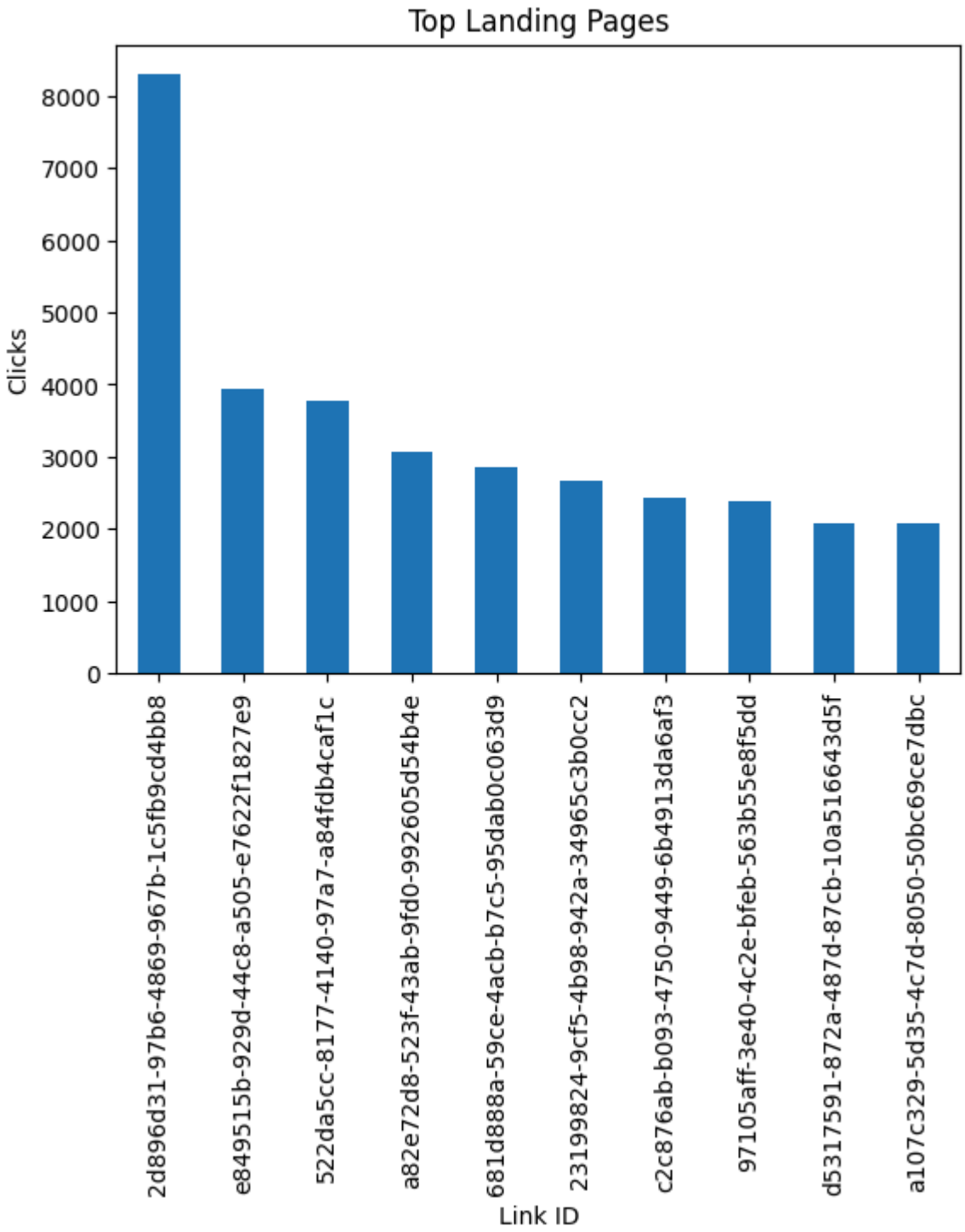
```
linkid
2d896d31-97b6-4869-967b-1c5fb9cd4bb8    8288
e849515b-929d-44c8-a505-e7622f1827e9    3950
522da5cc-8177-4140-97a7-a84fdb4caf1c    3786
a82e72d8-523f-43ab-9fd0-992605d54b4e    3059
681d888a-59ce-4acb-b7c5-95dab0c063d9    2865
23199824-9cf5-4b98-942a-34965c3b0cc2    2672
c2c876ab-b093-4750-9449-6b4913da6af3    2442
97105aff-3e40-4c2e-bfeb-563b55e8f5dd    2390
d5317591-872a-487d-87cb-10a516643d5f    2074
a107c329-5d35-4c7d-8050-50bc69ce7dbc    2072
Name: count, dtype: int64
```

In [13]:

```
top_pages.plot(kind='bar')

plt.title("Top Landing Pages")
plt.xlabel("Link ID")
plt.ylabel("Clicks")

plt.show()
```



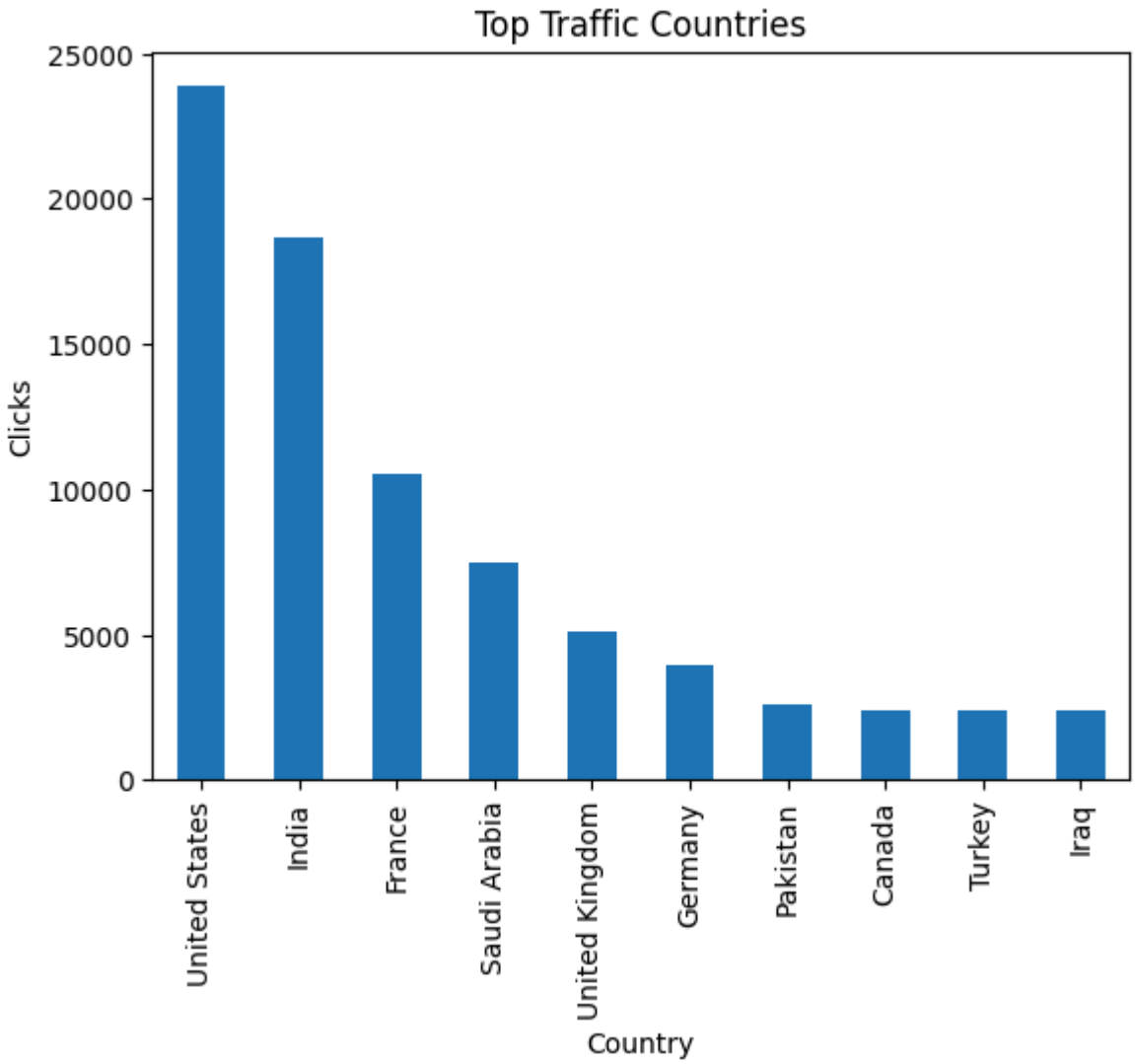
F. Top Countries(Traffic Source)

In [15]:
```
country_traffic = df['country'].value_counts().head(10)

country_traffic.plot(kind='bar')

plt.title("Top Traffic Countries")
plt.xlabel("Country")
plt.ylabel("Clicks")

plt.show()
```



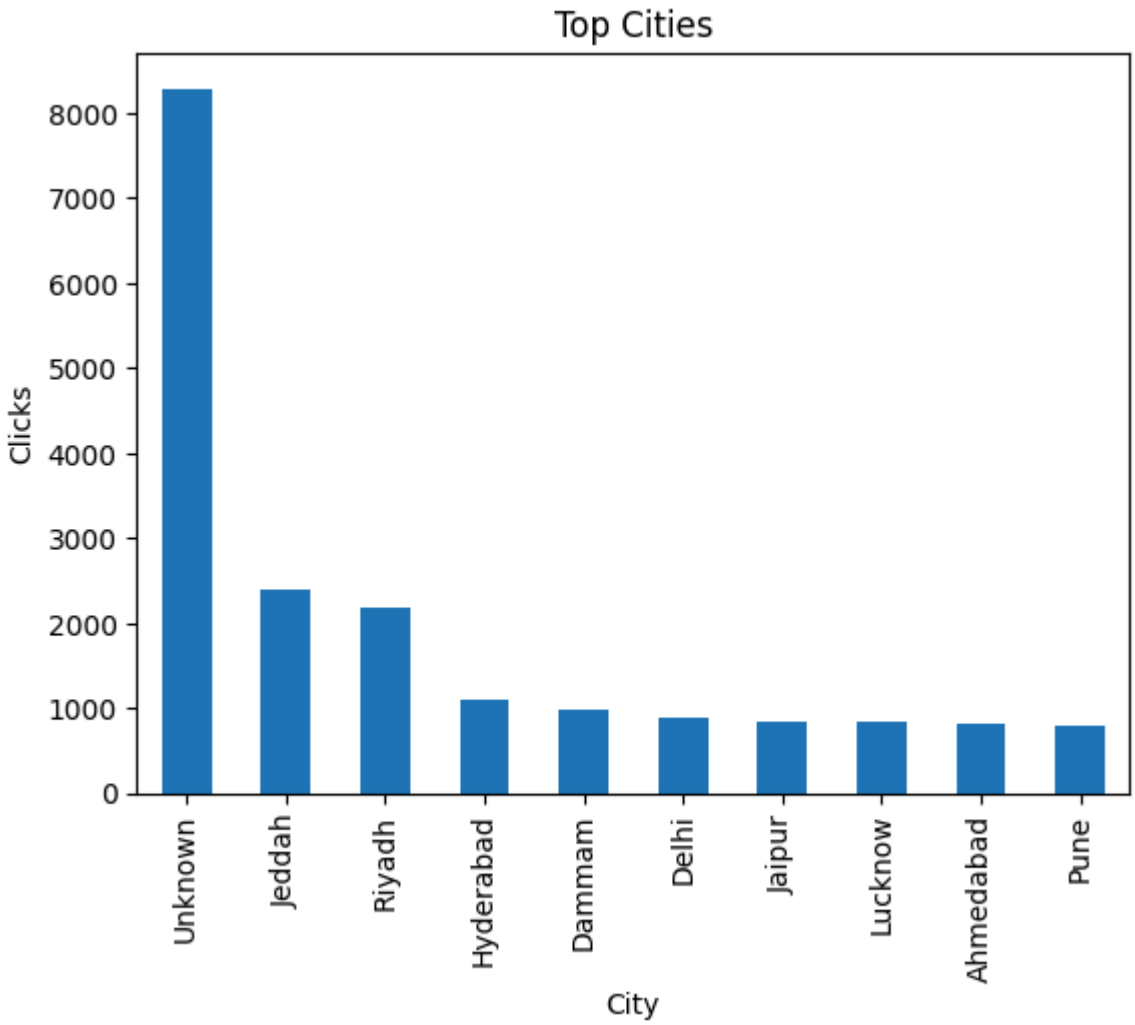
G. City Level Traffic

In [16]:
```
city_traffic = df['city'].value_counts().head(10)

city_traffic.plot(kind='bar')

plt.title("Top Cities")
plt.xlabel("City")
plt.ylabel("Clicks")

plt.show()
```



H. User Flow Visualization

In [17]:
```
flow = df.groupby(['artist','track']).size().reset_index(name='count')

flow.head()
```

Out[17]:

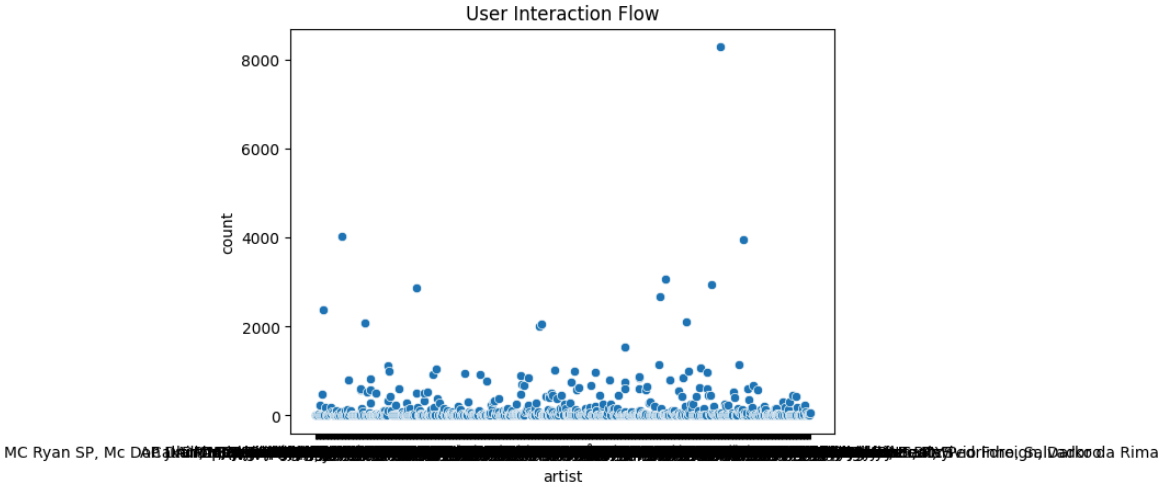
	artist	track	count
0	$NOT	Revenge	6
1	$ofaygo	Off the Map	7
2	22Gz	Fallen Blixkys	3
3	2KBABY	Great White	2
4	2KBABY, DDG	Zack & Cody (feat. DDG)	24

In [19]:
```
sns.scatterplot(data=flow, x='artist', y='count')

plt.title("User Interaction Flow")

plt.show()
```

```
C:\Users\pranj\AppData\Roaming\Python\Python311\site-packages\IPython\core\pylabt
ools.py:170: UserWarning: Glyph 27515 (\N{CJK UNIFIED IDEOGRAPH-6B7B}) missing fr
om font(s) DejaVu Sans.
  fig.canvas.print_figure(bytes_io, **kw)
C:\Users\pranj\AppData\Roaming\Python\Python311\site-packages\IPython\core\pylabt
ools.py:170: UserWarning: Glyph 20154 (\N{CJK UNIFIED IDEOGRAPH-4EBA}) missing fr
om font(s) DejaVu Sans.
  fig.canvas.print_figure(bytes_io, **kw)
```



I. Entry Pages

In [20]:
```
entry_pages = df['linkid'].value_counts().head(5)

print(entry_pages)
```

```
linkid
2d896d31-97b6-4869-967b-1c5fb9cd4bb8    8288
e849515b-929d-44c8-a505-e7622f1827e9    3950
522da5cc-8177-4140-97a7-a84fdb4caf1c    3786
a82e72d8-523f-43ab-9fd0-992605d54b4e    3059
681d888a-59ce-4acb-b7c5-95dab0c063d9    2865
Name: count, dtype: int64
```

J. Exit Pages

In [21]:
```
exit_pages = df.groupby('linkid').size().sort_values(ascending=False).head(5)

print(exit_pages)
```

```
linkid
2d896d31-97b6-4869-967b-1c5fb9cd4bb8    8288
e849515b-929d-44c8-a505-e7622f1827e9    3950
522da5cc-8177-4140-97a7-a84fdb4caf1c    3786
a82e72d8-523f-43ab-9fd0-992605d54b4e    3059
681d888a-59ce-4acb-b7c5-95dab0c063d9    2865
dtype: int64
```

Key Insights

1. Majority of traffic comes from top 3 countries.

2. Few landing pages receive most of the clicks.

3. Some users interact only once leading to high bounce rate.

4. Traffic is concentrated in few cities.
5. Certain tracks/artists generate higher engagement.

5 Optimization Recommendations

1️⃣ Improve Top Landing Pages

Optimize the most visited pages with clear CTA buttons.

2️⃣ Reduce Bounce Rate

Add:

engaging content

recommendations

related links

3️⃣ Personalize by Location

Since traffic comes from specific countries, create localized content.

4️⃣ Improve Internal Navigation

Guide users to next pages using:

recommended tracks

suggested artists

5️⃣ Mobile Optimization

Most traffic usually comes from mobile devices, so optimize page speed and UI.